



DATA PAPER OPEN ACCESS

Comprehensive Experimental Database for Validating CFD Simulations in Urban Wind Environment: Benchmark Cases Curated by the Architectural Institute of Japan

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Received: 8 January 2026 | **Revised:** 20 February 2026 | **Accepted:** 25 February 2026

Keywords: benchmark test | pedestrian-level wind | pollutant dispersion | wind environment | wind tunnel experiment

ABSTRACT

Reliable validation data are essential for ensuring the accuracy and applicability of computational fluid dynamics (CFD) simulations used to analyze urban wind environments. This data paper presents a comprehensive experimental database curated by a working group of the Architectural Institute of Japan (AIJ) to support CFD validation. The database includes 13 benchmark cases, ranging from simplified building geometries to complex urban configurations, with measurements of wind velocity, turbulence, pollutant (tracer) concentration, and temperature under various flow and thermal conditions. The experimental data were compiled from both existing and newly conducted wind-tunnel and field measurements, collected through collaborations among universities, research institutes, and industry. These cases provide a standardized and versatile basis for evaluating CFD model performance, facilitating uncertainty assessment, inter-model comparison, and best-practice development. Each case includes detailed information on experimental setups and measurement techniques to aid reproducibility and application. All datasets are publicly available through open data repositories and have already been widely used in numerous validation studies worldwide. This data paper aims to encourage continued use and broader dissemination of these benchmark cases, thereby enhancing the reliability and credibility of CFD predictions in urban wind and pollutant dispersion analyses.

1 | Introduction

Wind is one of the most significant factors affecting the urban environment. Wind-related environmental issues in urban areas can be categorized primarily into strong-wind and weak-wind problems. Strong wind problems primarily occur as pedestrian wind comfort, especially in the vicinity of high-rise buildings [1]. Insufficient wind can lead to environmental problems, such as air pollution and urban heat intensification [2]. Both types of problems arise from the interaction of atmospheric

boundary-layer flows with complex flow fields perturbed by buildings.

Wind tunnel (occasionally water tunnel) tests using scale models have been widely used to predict and analyze urban wind environments [3]. However, these methods have inherent limitations in capturing the detailed three-dimensional (3-D) spatial distribution of wind velocity, which is crucial for investigating environmental wind problems. Another limitation of wind tunnel experiments is the constraint of similarity requirements,

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which pose a major challenge when addressing environmental issues that involve not only wind velocity but also multiple physical processes, including thermal factors.

These limitations were greatly mitigated by the advent of computational fluid dynamics (CFD) for wind engineering, referred to as computational wind engineering (CWE) [4]. In the field of wind engineering, CWE research has been conducted systematically since the 1980s, and its impact on practice has been particularly significant for environmental issues compared with structural issues [5]. However, early knowledge of using CFD to predict the wind environment was largely accumulated through individual studies and was neither integrated nor standardized within the research community. Consequently, unreliable CFD predictions were occasionally reported, often because of the use of inappropriate conditions or methods.

Under these circumstances, in 1997, a working group was organized at the Architectural Institute of Japan (AIJ). The mission of this working group was to summarize key considerations and clarify the application range of CFD for predicting the pedestrian-level wind environments, providing a guidebook. Researchers from universities and private companies participated in this collaborative effort. Within the project, original benchmark tests were designed to investigate the influence of various computational conditions on different flow fields, and most of the wind tunnel experiments were conducted by the group members. The tests covered scenarios from a simple model to a complex urban configuration. Using the established databases, extensive sensitivity analyses were performed, including evaluations of turbulence models, grid discretization, computational domain size, and the representation of surrounding buildings [6–8]. In 2007, a CFD guidebook for predicting the pedestrian wind environment around buildings, written in Japanese, was published by the AIJ [9]. This guidebook presented the fundamentals of wind engineering and CFD, together with best-practice guidelines proposed by the working group [10]. It also included computational results and corresponding calculation conditions for the benchmark tests [11].

The AIJ guidebook, published in 2007, and the accompanying guidelines assumed that the Reynolds-averaged Navier–Stokes (RANS) model would be applied to pedestrian wind environments around high-rise buildings. However, the application scope of CFD has since expanded [12], supported by rapid advances in computing power. Given this background, the AIJ group expanded the scope of their guidebook to include the use of large-eddy simulation (LES) and the application of CFD to pollutant and heat dispersion analyses. In 2020, a revised AIJ guidebook was published in Japanese [13]. The cases in the experimental database for the benchmark tests were also extended by adding new cases that include pollutant and heat dispersions around buildings. Furthermore, building on these developments, members of the AIJ working group proposed specific guidelines for the application of LES to the pedestrian wind environment [14].

Although the AIJ working group has continuously provided and updated information and data regarding these benchmark tests through guidebooks, academic papers, and the institute's

website, a single document offering a unified and comprehensive explanation of the benchmark cases has been lacking. In addition, the data have not been fully utilized owing to challenges such as obsolete data formats, disorganized information, and a failure to align with recent trends in data provision, particularly the use of online repositories. To address these issues, this data paper systematically compiles the experimental conditions and measurement data for the benchmark tests developed by the AIJ working group. Reliable experimental data are essential for conducting robust validation and uncertainty assessment of CFD simulations and other numerical methods. These experimental databases are expected to enable more reliable numerical analyses of urban wind environments.

2 | Outlines of Benchmark Cases and Database

The validation database comprises 13 cases (Case A—Case M), which are summarized in Table 1. Overviews of the experimental conditions for each case are provided in the sub-sections of Section 3. When an original paper providing a more detailed description of the experiment exists, it is identified in the relevant sub-section, and both the original paper and the present paper should be cited when using the experimental data. In addition, representative studies that used the experimental data for validation are cited as references in the following sections.

Regarding data quality, although providing a unified statistical error analysis across all cases is not feasible given their diverse origins over several decades, specific information on measurement uncertainty is included for Cases F, H, J, and M on the basis of their original and relevant reports. For more detailed assessments of experimental errors, users are encouraged to consult the original primary publications cited in each sub-section.

3 | Details of Benchmark Cases

3.1 | Case A – Flow Around a 1:1:2 Shape Building Model

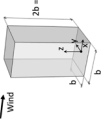
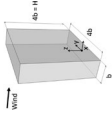
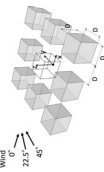
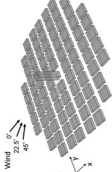
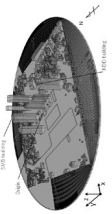
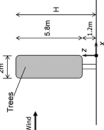
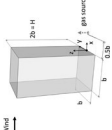
3.1.1 | Overview and Experimental Conditions

The velocity distributions were measured for the flow around a 1:1:2-shaped building model in a turbulent boundary layer. The model geometry is shown in Figure 1. The experimental results were originally reported by Meng and Hibi [15].

The experiment was conducted in a closed-circuit wind tunnel at the Technical Research Institute of Shimizu Corporation. The length of the measurement section of the wind tunnel was 7 m, and the height and width were 0.9 and 1.1 m, respectively. The atmospheric turbulent boundary layer corresponding to category IV (power-law exponent $\alpha=0.27$) in the AIJ recommendations for Loads on buildings [16] was created using spires, fences, and roughness blocks.

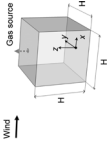
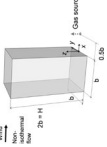
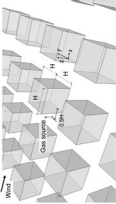
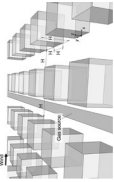
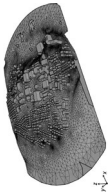
The model building had a width and depth of $b=0.08$ m and a height of $H=0.16$ m. The average inflow velocity at the building height U_H was 5.1 m/s. The Reynolds number (Re) on the basis of U_H and H was 2.4×10^4 .

TABLE 1 | Outlines of benchmark cases and database.

Case	Geometry and condition	Figure of the model	Wind direction	Available variables										
				U	V	W	V _S	σ _u	σ _v	σ _w	k	C	σ _c	Θ
A	Flow around a 1:1:2 shape building model (Isolated building)		0°	✓	✓	✓	✓	✓	✓	✓	✓			
B	Flow around a 1:4:4 shape building model (Isolated building)		0°	✓	✓	✓		✓	✓	✓	✓			
C	Flow within simple building blocks (Idealized urban area)		0°, 22.5°, 45°				✓							
D	Flow around a high-rise building in city blocks (Idealized urban area)		0°, 22.5°, 45°	✓	✓	✓	✓	✓	✓	✓	✓			
E	Flow within an actual urban area in Niigata (Actual urban area, H _{max} = 60 m)		16 directions (22.5° pitch)				✓							
F	Flow within an actual urban area in Shinjuku (Actual urban area, H _{max} = 210 m)		16 directions (22.5° pitch)				✓							
G	Flow behind a tree belt (Vegetation)		0°				✓				✓			
H	Flow and dispersion around a 1:1:2 shape building model (Isolated building)		0°	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓

(Continues)

TABLE 1 | (Continued)

Case	Geometry and condition	Figure of the model	Wind direction	Available variables										
				U	V	W	V _S	σ _u	σ _v	σ _w	k	C	σ _c	Θ
I	Flow and dispersion around a cubic building model (Isolated building)		0°	✓	✓	✓					✓			
J	Flow and dispersion around a 1:1:2 shape building model under non-isothermal conditions (Isolated building)		0°	✓	✓	✓		✓ ^a	✓ ^a	✓ ^a	✓	✓	✓	✓ ^b
K	Flow and dispersion within cube arrays with a point source (Idealized urban area)		0°	✓	✓	✓		✓	✓	✓	✓	✓	✓	
L	Flow and dispersion around cube arrays with a line-source under non-isothermal conditions (Idealized urban area)		0°	✓	✓	✓		✓ ^a	✓ ^a	✓ ^a	✓	✓	✓ ^b	✓ ^b
M	Flow and dispersion within the TPU campus (Actual urban area, H _{max} = 41 m)		0°	✓	✓	✓		✓	✓	✓				

Abbreviations: C = Time-averaged concentration, H_{max} = Maximum building height (full scale), k = Turbulence kinetic energy, U = Time-averaged velocity along the streamwise direction (x), V = Time-averaged velocity along the lateral direction (y), V_S = Time-averaged scalar velocity (speed), W = Time-averaged velocity along the vertical direction (z), Θ = Time-averaged temperature, σ_c = Standard deviation of concentration fluctuation, σ_u = Standard deviation of velocity fluctuation along the streamwise direction (x), σ_v = Standard deviation of velocity fluctuation along the lateral direction (y), σ_w = Standard deviation of velocity fluctuation along the vertical direction (z), σ_θ = Standard deviation of temperature fluctuation.
^aShear stress components are also available.
^bTurbulent diffusion fluxes are also available.

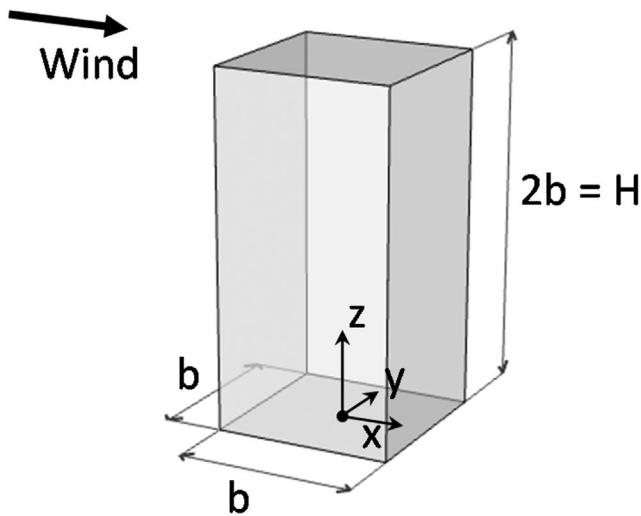


FIGURE 1 | Geometry of building model in Case A.

The measurements were performed on a vertical section of the building center ($y/b=0$) and on horizontal planes at $1/16$ ($z/b=0.125$) and $10/16$ ($z/b=1.25$) of the building height. All three components of wind velocity were measured using a hot-wire anemometer with a split-fiber probe (SFP). The cutoff frequency of the low-pass filter was set at 500 Hz. The measurement was made at a frequency of 1000 Hz for 60 s at each point.

3.1.2 | Data and Related Works

The experimental data can be accessed via the open repository Zenodo [17]. When using these data, we recommend also citing the original experimental paper [15]. The data have been used for validation in numerous studies [6, 18–35].

3.2 | Case B—Flow Around a 1:4:4 Shape Building Model

3.2.1 | Overview and Experimental Conditions

The velocity distribution was measured for the flow around a building model with a 1:4:4 (depth: width: height) shape placed in a boundary-layer flow. The geometry of the model is shown in Figure 2.

The measurements were performed using the closed-circuit wind tunnel at the Niigata Institute of Technology. The atmospheric turbulent boundary layer was generated using spires, fences, and blocks, following a power law with an exponent of 0.25.

The model had dimensions $b=0.05$ m (b : building depth) and $H=0.2$ m ($=4.0b$) (H : building height). The average approaching flow velocity at the building height U_H was 5.1 m/s. The Reynolds number, on the basis of U_H and H , was 7.2×10^4 .

The measurements were performed on a vertical section of the building center ($y/b=0$) and on horizontal planes at $1/16$ of the building height ($z/b=0.25$). The time-averaged and fluctuating

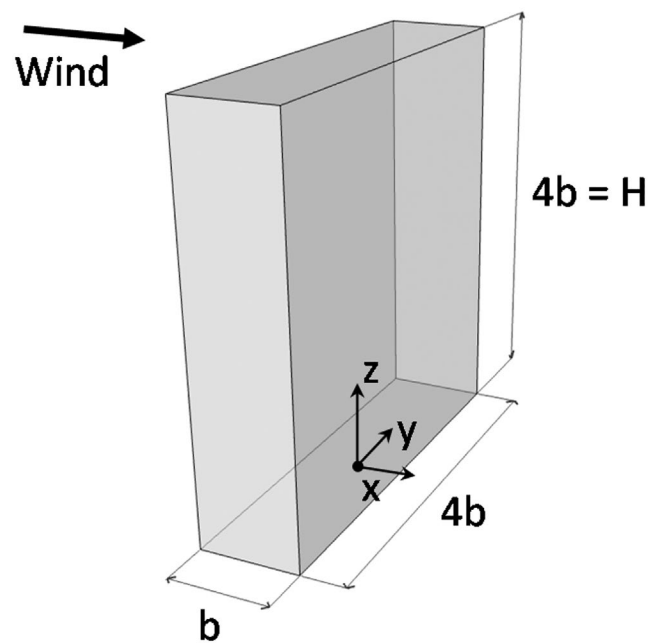


FIGURE 2 | Geometry of building model in Case B.

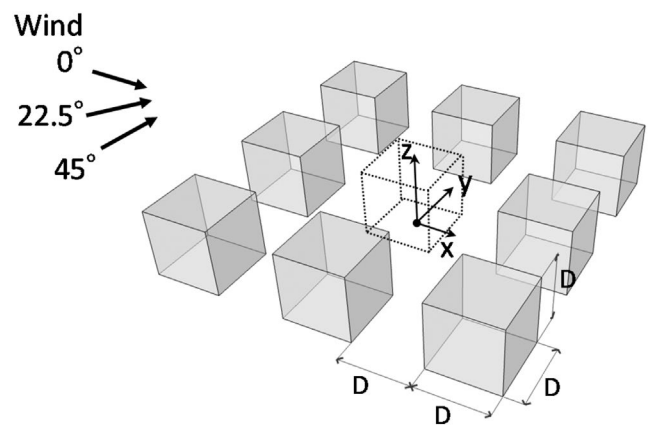


FIGURE 3 | Geometry of building blocks in Case C.

components of wind velocity were measured using an SFP in each of the three directions. The measurement was made at a frequency of 100 Hz for 60 s at each point.

3.2.2 | Data and Related Works

The experimental data can be accessed via Zenodo [36]. When using these data, we recommend also citing the original experimental paper [37]. The data have been used for validation in [38].

3.3 | Case C—Flow Within Simple Building Blocks

3.3.1 | Overview and Experimental Conditions

The velocity distribution was measured for the flow around a group of nine cuboid buildings installed in a boundary-layer flow (one central building and eight surrounding buildings). The model geometry is shown in Figure 3. The building model

was a cuboid with a side length $D=0.2$ m. Table 2 lists the cases investigated. Three shapes were used for the central building: [no building], $[D(x) \times D(y) \times D(z)]$, and $[D(x) \times D(y) \times 2D(z)]$. Three wind directions were used for each case: 0° , 22.5° , and 45° .

The wind tunnel experiment was performed using a Gettungen-type wind tunnel (measurement section: $1.8 \text{ m} \times 3.0 \text{ m}$) at the Technical Center of Fujita Corporation [39]. The exponent of the vertical profile of the approaching flow was approximately 0.28, and the turbulence intensity at height (D) was approximately 0.2. The average velocity of the approaching flow at the surrounding building height $H (= D)$ was set to $U_H = 3.65 \text{ m/s}$. The Reynolds number on the basis of U_H and H was 5.2×10^4 . The scalar wind velocity was measured at $0.1H$ height using thermistor anemometry with a sampling interval of 1 Hz and a sampling time of 30 s.

Since thermistor anemometers measure omnidirectional scalar wind speeds, careful attention must be paid to the consistency of the definition of “mean wind speed” when comparing experimental data with CFD results, which typically evaluate mean vector velocities [7, 23]. In light of this, particularly when using RANS ($k-\epsilon$ type) CFD analysis, a simple and conservative

correction that accounts for turbulent kinetic energy is sometimes applied to bridge the gap between scalar and vector mean velocities [11].

3.3.2 | Data and Related Works

The experimental data can be accessed via Zenodo [40]. When using these data, we recommend also citing the original experimental paper [39]. The data have been used for validation in many studies [41–49].

3.4 | Case D—Flow Around a High-Rise Building in City Blocks

3.4.1 | Overview and Experimental Conditions

A high-rise building in city blocks is modeled using a 1:1:4-shaped central building ($0.25H \times 0.25H \times H$). The representative height H is 0.25 m in model scale, corresponding to 100 m in full scale at a 1/400 scale ratio. This building is surrounded by uniformly arranged city blocks, each with a plane shape of $0.4H \times 0.4H$ and a height of $0.1H$. Each block was composed of two $0.1H$ -wide roads and additional roads with widths of $0.2H$ and $0.3H$. The model geometry is shown in Figure 4. Three wind directions (0° , 22.5° , and 45°) were considered for each case.

The wind tunnel experiment was conducted at the Niigata Institute of Technology. The atmospheric turbulent boundary layer was created using spires, fences, and blocks, following a power law with an exponent of 0.25. The inflow velocity at the central building height (H), U_H , was 6.65 m/s. The Reynolds number on the basis of U_H and H was 1.1×10^5 .

All measurements were taken at a height of 5 mm above the ground in model scale (equivalent to 2 m in full scale). The scalar wind velocity was measured using a thermistor anemometer (Exp_T). Additionally, the three components of the mean velocity and their fluctuations were measured by SFP near the central

TABLE 2 | Experimental cases for Case C.

No.	Center building	Wind direction
1	No building	0°
2		22.5°
3		45°
4	$D \times D \times D$	0°
5		22.5°
6		45°
7	$D \times D \times 2D$	0°
8		22.5°
9		45°

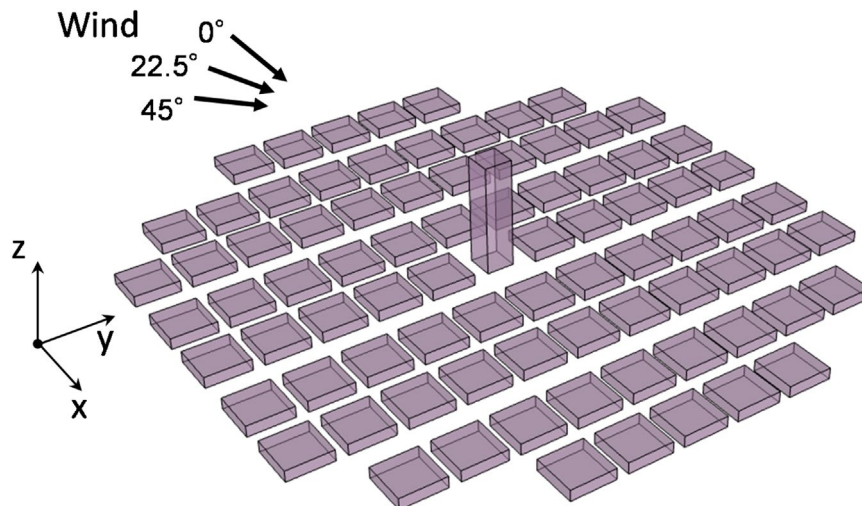


FIGURE 4 | Geometry of a high-rise building and city blocks in Case D.

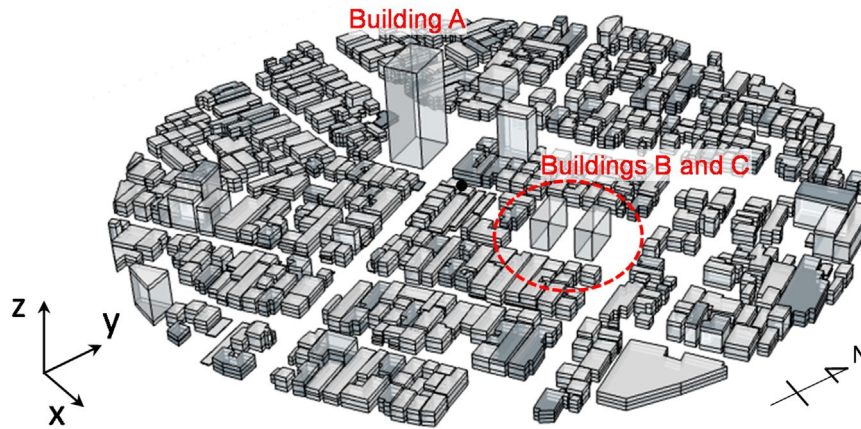


FIGURE 5 | Geometry of building complexes in Case E (after construction of planned building).

building for the 0° wind direction (Exp_S). Measurements using the SFP were conducted at a frequency of 100 Hz for 60 s at each point.

3.4.2 | Data and Related Works

The experimental data can be accessed via Zenodo [50]. The data have been used for validation in [34, 51, 52].

3.5 | Case E—Flow Within an Actual Urban Area in Niigata

3.5.1 | Overview and Experimental Conditions

The building complex represents a district with a dense concentration of two-story low-rise buildings on a flat topography near the mouth of the Shinano River in Niigata City. Figure 5 shows the geometry of the entire building complex. The experiments were conducted in two cases, one before and one after the construction of a 60 m high-rise building (Building A) and two 18 m mid-rise buildings (Buildings B and C). Building A faces relatively wide streets on the north, east, and south sides of its premises. The highest existing building (at a height of 30 m) in the target block stands approximately 40 m to the northeast of Building A. Buildings B and C face relatively wide streets on the north and east sides of their premises, with narrow lanes on their south and west sides. They are surrounded by two-story low-rise buildings.

The wind tunnel experiment was performed using the boundary-layer wind tunnel at the Niigata Institute of Technology. The wind tunnel section is 1.8 m (width) $\times$ 1.8 m (height). The diameter of the turntable is 1.6 m (corresponding to 400 m in actual scale). The vertical profile of the approaching wind velocity followed a power law with an exponent of 0.25. The inflow wind velocity at 1.0 m, the boundary layer height of the wind tunnel, was 7.8 m/s.

The scale of the experimental model was 1/250. The reference height z_R was set at 15.9 m (in full scale), at which wind velocity is measured at the nearest meteorological office. The Reynolds

number on the basis of z_R in the model scale, and the inflow wind velocity at that height (4.0 m/s) was 1.7×10^4 . The scalar wind velocity was measured using thermistor anemometry at a height of 2 m above ground level in full scale. The measurements were made at a frequency of 100 Hz for 60 s at each point.

3.5.2 | Data and Related Works

The experimental data can be accessed via Zenodo [53]. The data have been used for validation in [54–58].

3.6 | Case F—Flow Within an Actual Urban Area in Shinjuku

3.6.1 | Overview

The building complex represents the Shinjuku sub-city center at an early stage of development. Detailed wind tunnel tests and field measurements were conducted at the construction stage with the cooperation of various research institutes [59]. Figure 6 illustrates the geometry of the building complexes reproducing the situation at that time.

3.6.2 | Overview and Experimental Conditions

During the construction of the high-rise building group in the Shinjuku subcenter area of Tokyo, Japan, field observations of near-ground wind conditions were conducted from December 1975 to November 1983 [59]. In addition, several wind-tunnel experiments reproducing the building configurations at different construction stages were conducted [59]. This data set provides field observation data collected between December 1977 and November 1978, together with wind-tunnel experimental results corresponding to the 1978 building configuration.

The wind tunnel experiment was conducted in a wind tunnel at Shimizu Corporation. The cross-section of the wind tunnel was 1.55 m along the horizontal (y) direction and 2.2 m along the vertical (z) direction. The vertical profile of the mean inflow wind velocity followed a power law with an exponent of 1/4. The

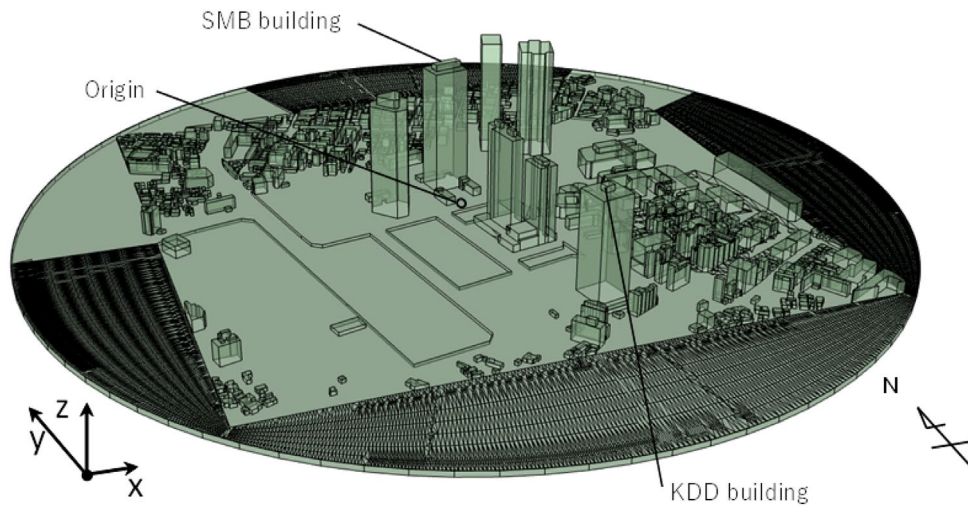


FIGURE 6 | Geometry of building complexes in Case F.

boundary layer height was assumed to be approximately 500 m, and the experimental wind speed was 10 m/s at that height. The model scale was set to 1/800, and an area with a diameter of 1600 m was modeled. Wind velocity was measured using a thermistor anemometer with an averaging time of 30 s at heights equivalent to 5, 9, and 20 m above ground level in full scale.

In contrast, for the field observations, a three-cup anemometer with a wind vane was employed, and the measurement height ranged from 3 to 9 m above the ground level, depending on the measurement point. The data obtained from December 1977 to November 1978, corresponding to the 1978 building configuration reproduced in the wind tunnel experiment, are provided. Average wind velocity ratios, normalized by a reference wind velocity, were obtained for each wind direction, under the condition that the reference wind velocity exceeds 5 m/s. For wind directions NW, NNW, N, NNE, NE, the reference velocity was applied as the wind velocity measured on the rooftop of the SMB building (rooftop elevation G.L. + 210 m, anemometer height G.L. + 237 m), whereas for other wind directions, the reference velocity was that of the KDD building (rooftop elevation G.L. + 165 m, anemometer height G.L. + 187 m). For the field observations, the standard deviations of the mean wind speed ratios at each observation point are provided in the dataset to indicate measurement variation.

3.6.3 | Data and Related Works

The experimental data can be accessed via Zenodo [60]. When using these data, we recommend also citing the original paper [59].

3.7 | Case G—Flow Behind a Tree Belt

3.7.1 | Overview and Measurement Conditions

Wind velocity distributions were measured around the Tsuiji Pines, a screen-like windbreak composed of black pines (*Pinus thunbergii*) located in the Izumo region of Shimane Prefecture, Japan. The dimensions of the Tsuiji Pines are approximately 2 m

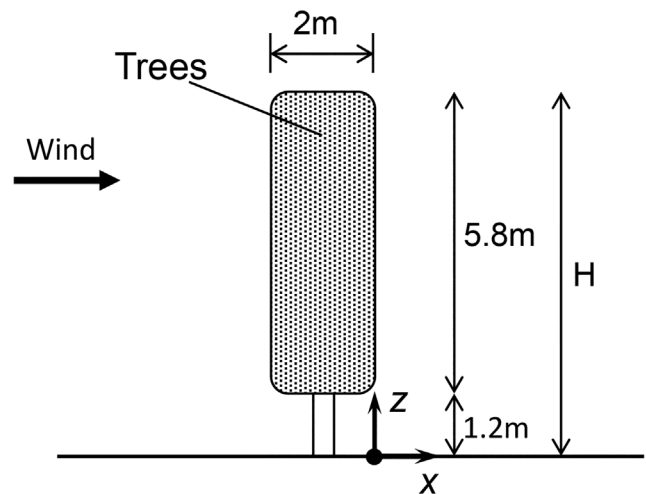


FIGURE 7 | Geometry of a two-dimensional pine tree in Case G.

(x) $\times$ 74 m (y) $\times$ 7 m (z). The measurements were conducted after the pruning state (shading rate: 81%). The geometry of the two-dimensional tree model is shown in Figure 7. The measurement results were originally reported by Kurotani et al. [61].

The field measurements were conducted at the west end of an elementary school playground. The vertical profile of the mean wind velocity of the approaching flow corresponded well to the power law with an exponent of 0.22.

To measure the distributions of the time-averaged velocity and turbulent kinetic energy on the leeward side of the trees, a movable observation tower equipped with three-dimensional ultrasonic anemometers at four heights, 1.5, 3, 4.5, and 6 m, was utilized. The measurement lines were $0.25H$, $0.5H$, $1.0H$, $2.0H$, $3.0H$, $4.0H$, and $5.0H$ behind the tree, where H is the tree height (7 m). Measurements were taken for 10 min for time averaging, with a sampling frequency of 5 Hz at each line. The wind velocities at each line were normalized by the mean reference wind speed measured at a height of 7 m, corresponding to the tree height, during each measurement period.

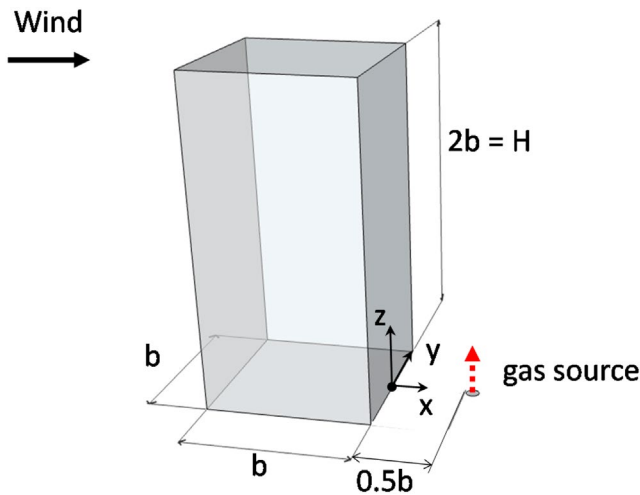


FIGURE 8 | Geometry of building model in Case H.

3.7.2 | Data and Related Works

The measurement data can be accessed via Zenodo [62]. When using these data, we also recommend citing the original paper on the field measurement [61]. The data have been used for validation in [63–68].

3.8 | Case H—Flow and Dispersion Around a 1:1:2 Shape Building Model

3.8.1 | Overview and Experimental Conditions

The flow and gas concentration fields were measured around a 1:1:2 (width: depth: height) shape building model placed in a turbulent boundary layer with tracer gas released from an outlet located on the downwind-side floor of the building. Figure 8 shows the building geometry and the arrangement of the gas outlets used in the experiment. Although results for the same configuration have been published in the wind tunnel experiment database provided by Tokyo Polytechnic University (TPU) [69], the measurements were reformed using the same configuration for this database.

The experiments were conducted in an Eiffel-type boundary-layer wind tunnel at TPU. The cross-section of the wind tunnel was 1.0 m along the height (z) direction and 1.2 m along the width (y) direction.

The building model has dimensions $b = 0.1$ m (b : building width), $H = 0.2$ m ($= 2.0b$) (H : building height). The wind speed at the building height $U_H = 3.23$ m/s, and the Reynolds number obtained from U_H and H was approximately 4.3×10^4 . A tracer gas outlet with a diameter of 4 mm was installed on the floor $0.25H$ away from the downwind wall of the building. Ethylene (C_2H_4) at 100% concentration was used as the tracer gas, and a volumetric flow rate of $q = 5.83 \times 10^{-6}$ m³/s was released from the outlet. The reference concentration $c_0 (= q / U_H H^2)$ used for normalization was 45.2 ppm.

The measurements were performed on a vertical section of the building center ($y/H = 0$) and on horizontal sections at z/H

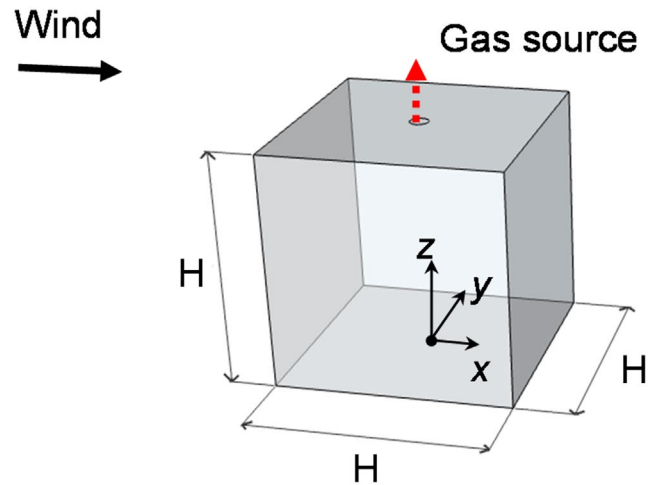


FIGURE 9 | Geometry of building model and gas source in Case I.

$= 0.0625, 0.25$, and 0.5 . The mean values and standard deviations of the three velocity components were measured using the SFP. The mean and standard deviation of the concentrations were measured by the high-speed total hydrocarbon analyzer. The sampling frequency was set to 1000 Hz. These values were averaged for a period of 120 s to obtain statistically stationary values. The equipment's uncertainty was evaluated in accordance with ISO/IEC Guide 98–3:2008(E) [70], as detailed by Tanaka et al. [71].

3.8.2 | Data and Related Works

The experimental data can be accessed via Zenodo [72]. When using these data, we recommend also citing the work by Yoshie [73], which describes the details of the experiment. The data have been used for validation in [74–79].

3.9 | Case I—Flow and Dispersion Around a Cubic Building Model

3.9.1 | Overview and Experimental Conditions

The velocity and gas concentration distributions were measured around a cubic building model, with point-source gas release from the center of the roof. The geometry of the building model and the gas source is shown in Figure 9. This is one case of wind tunnel experiments conducted by the National Institute of Advanced Industrial Science and Technology (AIST) to validate the Diffusion Model with Computational Fluid Dynamics (DiMCFD).

The wind tunnel was 20 m long and had a cross-section of 3.0×2.0 m. The approaching flow followed a $1/7$ power-law profile. The model was a cube with a side of 100 mm ($= H$). The Reynolds number defined by H and U_H (1.7 m/s) was 1.1×10^4 . The gas outlet was circular, 6 mm in diameter, from which 10% ethane was released at 6.67×10^{-6} m³/s. The release moment ratio ($M = V_e / U_H$) was 0.14 (V_e denotes the release flow rate). The reference concentration $c_0 (= q / U_H H^2)$ used for normalization was 39.2 ppm.

The measurements were performed on a vertical section of the building center ($y/H = 0$) and a horizontal section at $z/H = 0.5$ for the velocity, and on a vertical section at $y/H = 0$ and a horizontal section at $z/H = 0$ for the concentration. Wind velocity was measured using a laser Doppler anemometer (Dantec Dynamics, Fiber flow, BSA F60 Flow Processor), with a sampling rate of 200 Hz and averaged over 240 s. Concentrations were measured using a hydrocarbon analyzer (Kimoto Electric, HADA-01), with a sampling rate of 1 Hz and averaged over 120 s.

3.9.2 | Data and Related Works

The experimental data can be accessed via Zenodo [80]. When using this data, we recommend also citing the original article that provided the experimental data [81]. The data have been used for validation in [82].

3.10 | Case J—Flow and Dispersion Around a 1:1:2 Shape Building Model Under Non-Isothermal Conditions

3.10.1 | Overview and Experimental Conditions

Wind velocity and concentration were measured around a 1:1:2 (width: depth: height) shape building model placed in a non-isothermal (unstable) turbulent boundary layer, with gas emissions from an outlet located on the downwind floor of the model. The geometry of the building model and the gas source is shown in Figure 10. Although results for the same configuration have been published in the wind tunnel experiment database provided by TPU [69], the measurements were performed individually on the same configuration for this database.

The experiment was conducted in a thermal stratified wind tunnel at TPU. The cross-section of the wind tunnel was 1.0 m along the height (z) direction and 1.2 m along the width (y) direction. The building model had dimensions $b = 0.08$ m (b : building width) and $H = 0.16$ m ($= 2.0 b$) (H : building height). The tracer gas was ethylene (C_2H_4) (5% concentration), which was released at $q = 9.17 \times 10^{-6}$ m³/s from a 5 mm diameter gas outlet located $0.5 b$ ($0.25 H$)

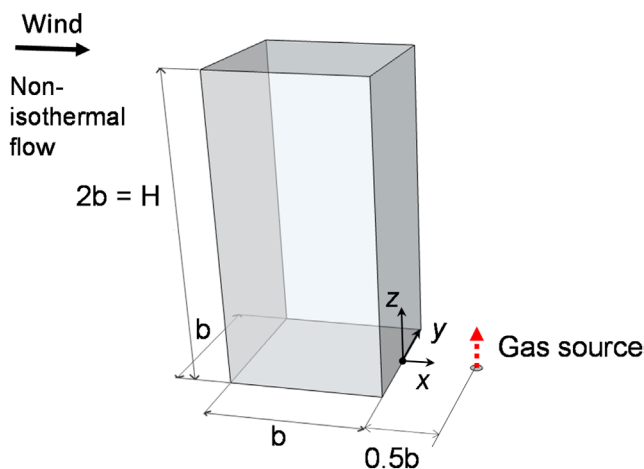


FIGURE 10 | Geometry of building model and gas source in Case J.

away from the leeward wall of the building. The average temperature of the gas is 30.4°C. The absolute difference between the temperatures of the approaching flow at height H and the floor, $\Delta\theta$, was 33.6°C. The Reynolds number obtained from H and U_H was approximately 1.6×10^4 . The bulk Richardson number on the basis of H , U_H , and $\Delta\theta$, and the representative temperature (average temperature from floor level to building height of the approaching flow) was approximately -0.086 .

The measurements were performed on a vertical section of the building center ($y/H = 0$) and on horizontal sections at $z/H = 0.025$ and 0.25 . The mean values and standard deviations of the three velocity components were measured using the SFP. The mean and standard deviation of the concentrations were measured by the high-speed hydrocarbon analyzer. The sampling frequency was set to 1000 Hz, and the measured data were time-averaged over 120 s to obtain statistically stationary values. A detailed technique for simultaneously measuring fluctuating velocity, temperature, and concentration in non-isothermal flow was reported by Yoshie et al. [83]. The equipment's uncertainty was evaluated in accordance with ISO/IEC Guide 98-3:2008(E) [70]. The legacy file in the dataset also provides the expanded uncertainty for each measured quantity to ensure the reliability of the data.

3.10.2 | Data and Related Works

The experimental data can be accessed via Zenodo [84]. When using these data, we recommend also citing the work by Yoshie [73], which provides detailed descriptions of the experiment. The data have been used for validation in [85].

3.11 | Case K—Flow and Dispersion Within Cube Arrays With a Point Source

3.11.1 | Overview and Experimental Conditions

The wind velocity and gas concentration were measured in an urban block model consisting of a group of cubes placed within the boundary layer, with tracer gas released from a point source located on the floor. The cube blocks were placed at equal intervals of 9 rows by 9 columns (81 cubes in total). The gas source was located between the second and third rows along the center line. The geometry of the model is shown in Figure 11.

The experiments were conducted in an Eiffel-type wind tunnel at TPU. The cross-section of the wind tunnel was 1.0 m along the height (z) direction and 1.2 m along the width (y) direction. The characteristics of the approaching flow were the same as those in Case H.

One side of the cube (H) was 0.06 m. The wind velocity at the height H of the approaching flow, U_H , was 2.33 m/s. The Reynolds number defined by H and U_H was 9.8×10^3 . The tracer gas was 100% C_2H_4 , released at $q = 3.6 \times 10^{-6}$ m³/s from a 5 mm diameter gas outlet.

The measurements were taken at approximately 30 points at eight different heights downwind of the line source. SFP was

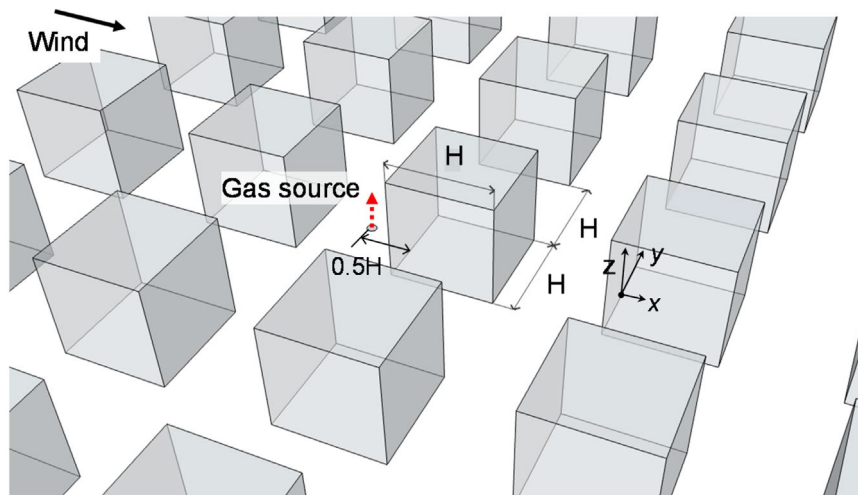


FIGURE 11 | Geometry of cube arrays and point source in Case K.

used to measure the wind velocity, and the high-speed hydrocarbon analyzer was used to measure concentrations. The sampling frequency was 1000 Hz, with a 200 Hz cutoff frequency applied to the low-pass filter. Measurements were conducted for 60 s to obtain statistical quantities.

3.11.2 | Data and Related Works

The experimental data can be accessed via Zenodo [86]. When using these data, we recommend also citing the work by Yoshie [73], which describes the details of the experiment. The data have been used for validation in [78, 87–92].

3.12 | Case L—Flow and Dispersion Around Cube Arrays With a Line-Source Under Non-Isothermal Conditions

3.12.1 | Overview and Experimental Conditions

The wind velocity, gas concentration, and temperature were measured simultaneously in a city block model consisting of a group of cubes placed in an unstable boundary layer, with pollutant dispersion emitted from a line source placed on the central floor. The city block model consists of 9 rows by 14 columns (126 cubes in total) and was aligned with a building coverage ratio of 25%. A gas line source was located perpendicular to the flow and centered between the buildings. The geometry of the model is shown in Figure 12. The experimental details are reported by Yoshie et al. [93]. Although results for the same configuration have been published in the wind tunnel experiment database provided by TPU [69], the measurements were performed individually using the same configuration for this database.

The experiment was conducted in a thermally stratified wind tunnel at TPU. The cross-section of the wind tunnel was 1.0 m along the height (z) direction and 1.2 m along the width (y) direction. Twenty-six aluminum angles, each 9 mm high and

1200 mm long, were placed at 200 mm intervals on the wind tunnel floor as roughness elements. A pipe for releasing tracer gas was installed on the floor at the center of the blocks (between the seventh and eighth rows from the tip). The block model floor was raised 10 mm above the wind tunnel floor for the burial of the release pipe, and the upwind wind tunnel floor was smoothly connected to the block model floor by a 1/7 slope taper.

One side of the cube (H) was 60 mm. The width and length of the gas line source were 37 and 1200 mm, respectively. The wind velocity of the approaching flow at H was $U_H = 1.34$ m/s, the airflow temperature at the same height $\Theta_H = 14.8^\circ\text{C}$, and the temperature difference between the floor temperature Θ_f and the temperature Θ_δ at the boundary layer height $\Delta\Theta$ was 30.3°C . The Richardson number determined from the mean air velocity gradient and mean temperature gradient along the vertical direction was approximately -0.1 at the half-height of the boundary layer. C_2H_4 was released from the source at a concentration of 100% at a flow rate of $8.33 \times 10^{-6} \text{ m}^3/\text{s}$ and at 35°C [94].

The measurement points were placed at 8 points between 4 and 150 mm in height at 27 locations on the road near the source, at 4 points between 70 and 150 mm in height at 14 locations on the block model, and 3 points between 4 and 30 mm in height at 24 locations in the surrounding area. SFP, a high-speed hydrocarbon analyzer, and a cold wire were used to measure wind velocity, concentration, and temperature, respectively. Three sensors were placed close to each other for measuring wind speed, concentration, and temperature simultaneously [83]. The sampling frequency was 1000 Hz, and the low-pass filter cutoff frequency was 200 Hz. A total of 60000 data points were acquired over a 60 s measurement period.

3.12.2 | Data and Related Works

The experimental data can be accessed via Zenodo [95]. When using these data, we recommend also citing the original experiment paper by Yoshie et al. [93]. The data have been used for validation in [79, 91, 94].

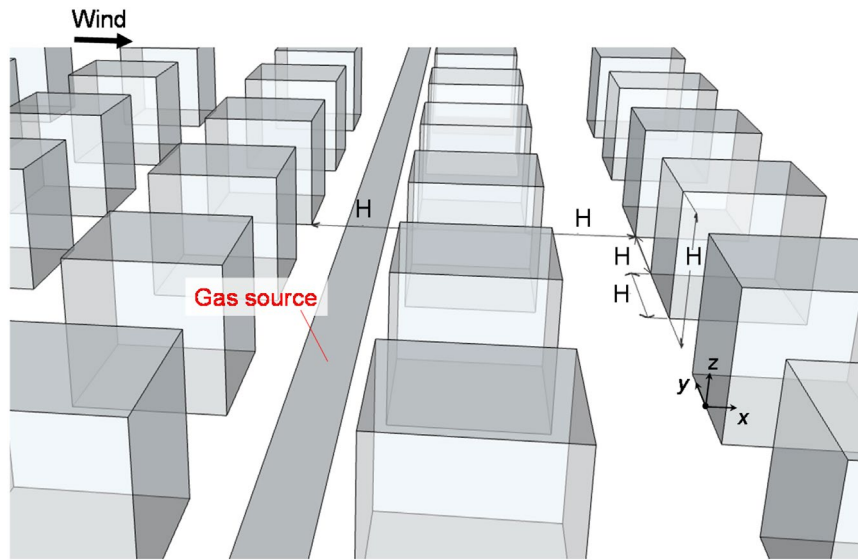


FIGURE 12 | Geometry of cube arrays and line source in Case L.

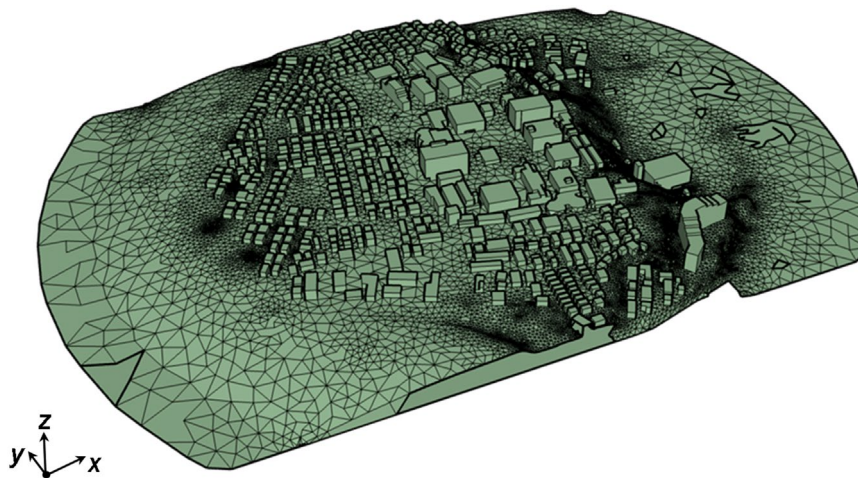


FIGURE 13 | Geometry of campus building model in Case M.

3.13 | Case M—Flow and Dispersion Within TPU Campus

3.13.1 | Overview and Experimental Conditions

The concentration of released pollutants was measured in a realistic building complex. This wind tunnel experiment was conducted to reproduce the field measurement campaign of tracer gas release at the Atsugi Campus of TPU [96]. A model of the area with a radius of approximately 300 m, centered on the tracer gas release point, was reproduced. Figure 13 shows the geometry of the scale model used for the wind tunnel experiment. The database also includes corresponding field measurement results [97]. As the dataset provides the results of each individual run, it enables an assessment of the reproducibility and uncertainty of the tracer gas dispersion results.

Wind tunnel experiments were conducted using an Eiffel-type boundary layer wind tunnel at TPU. The cross-section of the wind tunnel was 1.0 m along the height (z) direction and 1.2 m

along the width (y) direction. To account for the influence of the area's topography, a taper was placed upwind of the model. The approaching flow was generated in accordance with the vertical profile of the mean wind velocity and turbulent intensity corresponding to category IV (power-law exponent $\alpha=0.27$) in the AIJ recommendations for Loads on buildings [16].

The model scale was 1/600. The reference point, R , was defined as the rooftop of the main building, with a height (H_R) of 41 m in full scale and 0.068 m in model scale. The wind velocity U_R at R in the wind tunnel experiment was 4.08 m/s. C_2H_4 at 100% concentration was used as the tracer gas, which was released through a 6 mm diameter outlet at a volumetric flow rate of $2.50 \times 10^{-6} \text{ m}^3/\text{s}$ (exit velocity is 0.09 m/s). The reference concentration $c_0 = q / (U_R H_R^2)$ was $1.33 \times 10^2 \text{ ppm}$, where q is the volumetric flow rate of the tracer gas.

The trace-gas concentration and three wind velocity components were measured at 15 points, corresponding to those used in the field measurements. The high-speed hydrocarbon

analyzer was used to measure concentrations. The sampling frequency was 1000 Hz with a 200 Hz cutoff frequency applied to the low-pass filter, and measurements were conducted for 240 s at each point.

3.13.2 | Data and Related Works

The experimental data can be accessed via Zenodo [98]. When using these data, we recommend also citing the original papers for experiment and field measurements [96, 97]. The data have been used for validation in [92, 99, 100].

Acknowledgments

The authors would like to express their gratitude to the former members of the working group on CFD prediction of pedestrian wind environments around buildings, established within the Architectural Institute of Japan. In addition, we sincerely thank all collaborators who contributed to the experiments presented in this work. In particular, we thank Dr. Yasuo Kurotani (ex-Shimane University) for allowing the use of his experimental results. We also thank Ms. Rina Higashino (Institute of Science Tokyo) for her assistance in curating the benchmark data.

Funding

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All experimental datasets for the benchmark cases presented in this paper are openly available in the Zenodo repository. The specific links are as follows: Case A: <https://doi.org/10.5281/zenodo.15050147> [17]. Case B: <https://doi.org/10.5281/zenodo.15401455> [36]. Case C: <https://doi.org/10.5281/zenodo.15401791> [40]. Case D: <https://doi.org/10.5281/zenodo.15402668> [50]. Case E: <https://doi.org/10.5281/zenodo.15429017> [53]. Case F: <https://doi.org/10.5281/zenodo.15589621> [60]. Case G: <https://doi.org/10.5281/zenodo.15589666> [62]. Case H: <https://doi.org/10.5281/zenodo.15429871> [72]. Case I: <https://doi.org/10.5281/zenodo.15430018> [80]. Case J: <https://doi.org/10.5281/zenodo.15447887> [84]. Case K: <https://doi.org/10.5281/zenodo.15448050> [86]. Case L: <https://doi.org/10.5281/zenodo.15487637> [95]. Case M: <https://doi.org/10.5281/zenodo.15487857> [98]. Detailed information regarding the original experimental conditions and related publications is provided within the “Data and Related Works” sub-section of each case to ensure that the context of the data and its source are clearly understood.

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